4. The repulsive interaction between the different types of electron pairs is in the following order.

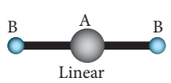
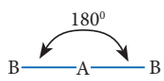
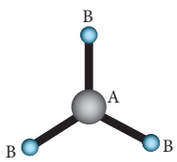
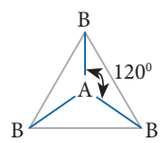
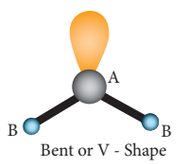
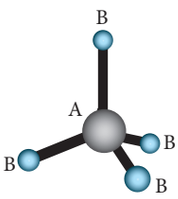
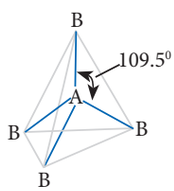
lp - lp > lp - bp> bp-bp

lp- lone pair ; bp- bond pair

The lone pair of electrons are localised only on the central atom and interacts with only one nucleus whereas the bond pairs are shared between two atoms and they interact with two nuclei. Because of this the lone pairs occupy more space and have greater repulsive power than the bond pairs in a molecule.

The following Table illustrates the shapes of molecules predicted by VSEPR theory. Consider a molecule AB_x where A is the central atom and x represents the number of atoms of B covalently bonded to the central atom A. The lone pairs present in the atoms are denoted as L.

Table 10. 6 Shapes of molecules predicted by VSEPR theory.

Number of electron Pairs	Molecule	No.of bond pairs	No.of. lone pairs	Shape	Molecular geometry	Examples
2	AB_2	2	-	 Linear	 Linear	$BeCl_2$, $HgCl_2$, CO_2, CS_2, HCN, BeF_2
3	AB_3	3	-	 Trigonal planar	 Trigonal planar	BF_3, BCl_3, NO_3^-, BF_3, CO_3^{2-}, HCHO
	AB_2L	2	1	 Bent or V - Shape	Bent or V - Shape	SO_2, O_3, $PbCl_2$, $SnBr_2$
4	AB_4	4	-	 Tetrahedral	 Tetrahedral	CH_4, CCl_4, CCl_2F_2, SO_4^{2-},ClO_4^-, NH_4^+

Unit 10 layout - jagan.indd 82 22-01-2021 12:33:12